DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Mathematics III   Topics in Analysis (3TN)

Normed Spaces

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0.1 Introduction

In this lecture, we study some particular metric spaces (X, d) where the set X is a vector space and the distance d is induced by an another entity called a norm.

Notice that the material presented in this lecture, will be very useful for your future studies in Mathematics. For instance, in MAM4000W, Honours in Pure Mathematics (first and second semester).

0.2 General notions on vector spaces (Revision)

Let $\mathbb{K}$ denote either the field of real numbers $\mathbb{R}$ or the field of complex numbers $\mathbb{C}$.

Definition 0.1 (Vector space)
A vector (or linear) space (over the field $\mathbb{K}$) is a non-empty set V with an addition operation between its elements (i.e., a mapping $(u, v) \to u + v$ from $V \times V$ to V) and a multiplication operation of its elements by numbers in $\mathbb{K}$ (i.e., a mapping $(\alpha, u) \to \alpha u$ from $\mathbb{K} \times V$ to V), that satisfy the following properties called axioms of a vector space

(i) $u + v = v + u$ for every $u,\, v \in V$ (commutative law);

(ii) $(u + v) + w = u + (v + w)$ for every $u,\, v,\, w \in V$ (associative law);

(iii) there exists a unique element $0_V \in V$ such that $0_V + u = u$ for all $u \in V$, and 0_V will be denoted by 0 when there is no ambiguity (existence of the zero element);

(iv) for every $u \in V$ there exists a unique element $-u \in V$ such that $u + (-u) = 0_V$ (opposite element);

(v) $1u = u$ for any $u \in V$;

(vi) $\alpha(\beta u) = (\alpha\beta)u$ for any $u \in V$, any $\alpha,\, \beta \in \mathbb{K}$ (associative law for scalar multiplication);

(vii) $\alpha(u + v) = \alpha u + \alpha v$ for any $u,\, v \in V$, and any $\alpha \in \mathbb{K}$ (distributive laws).

The elements of a vector space are called *vectors*. The field $\mathbb{K}$ is called the *scalar field* and its elements are called *scalars*.
If $\mathbb{K} = \mathbb{R}$, then V is called *real vector space* while the space V is a *complex linear space* when $\mathbb{K} = \mathbb{C}$.

Example 0.2
(1) The set of real numbers $\mathbb{R}$ is a real vector space when the addition and scalar multiplication are the usual addition and multiplication of real numbers. Similarly, the set of complex numbers $\mathbb{C}$ is a complex vector space.

(2) Let Ω be an open subset of $\mathbb{R}^n$. Let $C(\Omega)$ be the set of all real-valued functions which are defined and continuous on Ω. For f, $g \in C(\Omega)$ we define $f+g$ to be the function which takes each $x \in \Omega$ to the number $f(x)+g(x)$; i.e., $(f+g)(x) := f(x)+g(x)$ for all $x \in \Omega$. Similarly, given $\lambda \in \mathbb{R}$ we define $(\lambda f)(x) := \lambda f(x)$ for every $x \in \Omega$.
Under the algebraic operations just defined, the set $C(\Omega)$ is clearly a real vector space. Similarly, $C(\overline{\Omega})$ is a real vector space.

Definition 0.3 (Vector subspace)
Let V be a vector space over $\mathbb{K} = \mathbb{C}$ or $\mathbb{R}$. We call *subspace* of V any subset S of V which is closed under the addition and scalar multiplication, i.e., for any u, $v \in S$ and any $\lambda \in \mathcal{R}$, we have $u+v \in S$ and $\lambda u \in V$.

- We say that S is a *proper* subspace of V if S is a subspace and $S \neq V$.
- We say that the vector space V is the direct sum of the two subspaces S_1 and S_2 if any element $v \in V$ can be written as

$$v = v_1 + v_2 \quad \text{with} \quad v_1 \in S_1 \quad \text{and} \quad v_2 \in S_2$$

  and the decomposition is unique. In symbols

$$\boxed{V = S_1 \oplus S_2 \quad \Leftrightarrow \quad \begin{cases} V = S_1 + S_2 \\ S_1 \cap S_2 = \{0_V\} \end{cases}}$$

Definition 0.4 (Linear combination) Let V be a linear space over $\mathbb{K}$.
• Let S be a non-empty subset of V. A vector v is said to be a *linear combination* of vectors in S if there is a finite set $\{v_1, v_2, \ldots, v_n\} \subset S$ and scalars $\alpha_1, \alpha_2, \ldots, \alpha_n \in \mathbb{K}$ such that $v = \sum_{i=1}^n \alpha_i v_i$. The set of all linear combinations of vectors in S will be denoted by Span (S). If Span $(S) = V$, we say that S spans V.

• We say that a system of n elements $\{v_1, v_2, \ldots, v_n\} \subset V$ is *linearly dependent* it there exist n scalars $\alpha_1, \alpha_2, \ldots, \alpha_n \in \mathbb{K}$, at least one being non-zero, such that $\sum_{i=1}^n \alpha_i v_i = 0_V$. Otherwise, the vectors $v_1, v_2, \ldots, v_n$ are said to be *linearly independent*, i.e.,

$$\boxed{\sum_{i=1}^{n} \alpha_i v_i = 0_V \quad \Rightarrow \quad \alpha_1 = \alpha_2 = \cdots \alpha_n = 0.}$$

• An arbitrary (non-empty) subset S of V is called linearly independent if any finite part of S consists of linearly independent elements.

Example 0.5
(i) $\mathbb{R}$ and $\mathbb{C}$ are real linear spaces of dimension 1 and 2 respectively. The canonical basis of $\mathbb{R}$ and $\mathbb{C}$ are $\{1\}$ and $\{1, i\}$ respectively.

(ii) The set $\mathcal{P}_k(\mathbb{R})$ of polynomials of degree less or equal to k, with real coefficients is real vector space of dimension $k+1$ with basis, the monomials: $1,\ x,\ x^2,\ \dots,\ x^k$.

(iii) Let $\mathcal{P}_2(\mathbb{R}^2)$ and $\mathcal{Q}_2(\mathbb{R}^2)$ be the vector spaces of polynomials in $\mathbb{R}^2$ of total and partial degree ≤ 2 respectively. $\mathcal{P}_2(\mathbb{R}^2)$ and $\mathcal{Q}_2(\mathbb{R}^2)$ have canonical basis

$$\{1, x, y, x^2, xy, y^2\} \qquad \text{and} \qquad \{1, x, y, xy, x^2, y^2, x^2y, xy^2, x^2y^2\}$$

respectively, and hence $\dim \mathcal{P}_2(\mathbb{R}^2) = 6$ while $\dim \mathcal{Q}_2(\mathbb{R}^2) = 9$.

0.2.1 Normed spaces

Definition 0.6 (Normed space)
Let V be a vector space over the field $\mathbb{K} = \mathbb{R}, \mathbb{C}$. A *norm* on V is a function $N : V \to \mathbb{R}$ that satisfies the following properties:

(N1) $N(u) \geq 0$ for all $u \in V$ and $N(u) = 0 \Leftrightarrow u = 0$;

(N2) $N(\alpha u) = |\alpha| N(u) \quad \forall u \in V, \forall \alpha \in \mathbb{K}$;

(N3) $N(u+v) \leq N(u) + N(v) \quad \forall u, v \in V$.

If only (N2) and (N3) are satisfied we say that N is a *semi-norm*. If N is a norm on V, it is customary to denote, for every $u \in V$, the number $N(u)$ by $\|u\|$. The pair $(V, \|\cdot\|)$ is called *normed space*. V is also a metric space with distance $d(u,v) := \|u - v\|$. The distance d is called *distance induced by the norm*.

Example 0.7
(i) For every $u,\ v \in V$, writing $u = v + (u - v)$ or $v = u + (v - v)$, it from from the property (N3) that

$$|\|u\| - \|v\|| \leq \|u - v\|;$$

hence the norm is a Lipschitz continuous function from $(V, \|\cdot\|)$ into $\mathbb{R}$.

(ii) The following are norms on $\mathbb{R}^d$:

$$\|\boldsymbol{x}\|_p := \left(\sum_{i=1}^{d} |x_i|^p\right)^{1/p} \quad \text{for } 1 \leq p < \infty \text{ and } \|\boldsymbol{x}\|_\infty := \max_{1\leq i\leq d} |x_i| \quad \forall \boldsymbol{x} = (x_1, x_2, \cdots, x_d) \in \mathbb{R}^d.$$

(iii) Let $\Omega \subset \mathbb{R}^d$ be a bounded open set. For each $f \in C(\overline{\Omega})$, set

$$\|f\|_\infty := \sup_{x\in\overline{\Omega}} |f(x)|.$$

It is easy to see that $\|\cdot\|_\infty$ has the properties (N1)-(N3) and hence $(C(\overline{\Omega}), \|\cdot\|_\infty)$ is an infinite dimensional normed space. $\|\cdot\|_\infty$ is called the supremum norm.

We said in the definition 0.6 any normed space is also a metric space with the notion of distance induced by the norm. However, there distances which are not induced by any norm on the vector space. In the following theorem we will prove that any metric space (X, d) is isometric to some metric subspace of a normed space.

Theorem 0.8 *Every metric space (X, d) is isometric to a metric subspace of some normed space (V, N).*

Proof: Let $Y := \mathcal{B}(X, \mathbb{R})$ the vector space over the field $\mathbb{R}$ of bounded functions from X to $\mathbb{R}$, equipped with norm $\|\cdot\|_\infty$. Fix $a \in X$. For every $x \in X$, we define the function $\phi_x : X \to \mathbb{R}$ by

$$\boxed{\phi_x(y) := d(y, x) - d(y, a) \quad \forall y \in X}\,.$$

let us show that $\phi_x \in Y$. Using the triangle inequality property of the distance d, we obtain that

$$\phi_x(y)| = |d(y, x) - d(y, a)| \leq d(x, a) \quad \Rightarrow \quad \phi_x \in Y \text{ with } \|\phi_x\|_\infty \leq d(x, a)\,.$$

On the other hand, taking $y = a$, we get that $\|\phi_x\|_\infty \geq |\phi_x(a)| = d(x, a)$ and hence,

$$\boxed{\|\phi_x\|_\infty = d(x, a)}\,.$$

Now, consider the mapping $\Phi : X \to Y$ defined by

$$\boxed{\Phi(x) = \phi_x \quad \forall x \in X}\,.$$

Clearly the mapping Φ is well-defined. We claim that the mapping ϕ is an isometry, i.e.,

$$\boxed{\|\Phi(x) - \Phi(x')\|_\infty = d(x, x') \quad \forall x,\, x' \in X}\,. \qquad (*)$$

Let $y \in X$. We have

$$\begin{aligned} |(\Phi(x) - \Phi(x'))(y)| &= |\phi_x(y) - \phi_{x'}(y)| = |(d(y, x) - d(y, a)) - (d(y, x') - d(y, a)| \\ &= |d(y, x) - d(y, x')| \leq d(x, x')\,. \end{aligned}$$

Thus, $\|\Phi(x) - \Phi(x')\|_\infty \leq d(x, x')$. On the other hand, taking $y = x$, we get

$$\begin{aligned} \|\Phi(x) - \Phi(x')\|_\infty &\geq |(\Phi(x) - \Phi(x'))(x)| = |\phi_x(x) - \phi_{x'}(x)| \\ &= |(0 - d(x, a)) - (d(x, x') - d(x, a)| = d(x, x')\,. \end{aligned}$$

Therefore, the identity $(*)$ holds confirming that the mapping Φ is indeed an isometry from X to Y. So, the metric space (X, d) can be identified to a metric subspace of the normed space $(Y, \|\cdot\|_\infty)$. ■

Exercises 0.9 Let $n \in \mathbb{N}$ and let $\mathcal{P}_n$ be the set of polynomials $p(x) = a_k x^k + a_{k-1} x^{k-1} + \ldots + a_1 x + a_0$ with real coefficients a_i and $k \leq n$.
For any $p(x) = a_k x^k + a_{k-1} x^{k-1} + \ldots + a_1 x + a_0 \in \mathcal{P}_n$, set

$$\|p\| := \max\{|a_k|, \ldots, |a_0|\}\,.$$

Show that $\|\cdot\|$ is a norm on $\mathcal{P}_n$.

Definition 0.10 (Cauchy sequences - Converging sequences - Series)
Let $(V, \|\cdot\|)$ be a normed space over the field $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$ and let (v_n) be a sequence in V.

(a) We say that (v_n) is a Cauchy sequence in $(V, \|\cdot\|)$ if $\lim\limits_{n,m\to\infty} \|v_n - v_m\| = 0$, that is,

$$\boxed{\forall \varepsilon > 0 \quad \exists n_\varepsilon \in \mathbb{N} \text{ such that } \forall n,\, m \in \mathbb{N} \quad n, m \geq n_\varepsilon \quad \Rightarrow \quad \|v_n - v_m\| < \varepsilon}\,.$$

(b) We say that (v_n) is a converging sequence in $(V, \|\cdot\|)$ if there exists $v \in V$ such that $\lim_{n\to\infty} \|v_n - v\| = 0$, that is,

$$\boxed{\exists v \in V \quad \forall \varepsilon > 0 \quad \exists n_\varepsilon \in \mathbb{N} \text{ such that } \forall n \in \mathbb{N} \quad n \geq n_\varepsilon \quad \Rightarrow \quad \|v_n - v\| < \varepsilon}\,.$$

(c) We say that the series $\sum_{n=0}^{\infty} v_n$ converges in $(V, \|\cdot\|)$ if the sequence (s_m) of partial sums, i.e., $s_m := \sum_{n=0}^{m} v_n$ for every $m \in \mathbb{N}$, converges in $(V, \|\cdot\|)$. The limit s of the sequence (s_m) is denoted by $s = \sum_{n=0}^{\infty}$ and is called the sum of the series. Otherwise, we say that the series diverges.

If the series $\sum_{n=0}^{\infty} \|v_n\|$ converges, then we say that the series $\sum_{n=0}^{\infty} v_n$ converges absolutely.

Definition 0.11 (Banach spaces)
We say that the normed space $(V, \|\cdot\|)$ is a Banach space if (V, d) is a complete metric space, where d is the distance induced by the norm $\|\cdot\|$. That is, for every Cauchy sequence in (V, d) converges.

Remark 0.12 Notice that if $(V, \|\cdot\|$ is a banach space, then every absolutely converging series does converge. In fact, if the sequence (v_n) is such that $\sum_{n=0}^{\infty} \|v_n\| < \infty$, then

$$\|s_k - s_l\| \leq \sum_{n=k+1}^{l} \|v_n\| \to 0 \quad \text{for} \quad k < l \text{ and } k,\, l \to \infty\,.$$

Exercise 0.13 Let $\mathcal{B}(\mathbb{R}, \mathbb{R})$ be the real vector space of bounded functions from $\mathbb{R}$ to $\mathbb{R}$. Set

$$\boxed{\|f\|_\infty := \sup_{x\in\mathbb{R}} |f(x)| \qquad f \in \mathcal{B}(\mathbb{R}, \mathbb{R})}\,.$$

(a) Show that $\|\cdot\|_\infty$ is a norm on $\mathcal{B}(\mathbb{R}, \mathbb{R})$ and that $(\mathcal{B}(\mathbb{R}, \mathbb{R}), \|\cdot\|_\infty)$ is a Banach space. (Hint: you just need to mimic the proof in your 3MS notes on $(\mathcal{C}(X, Y), d_\infty)$ a complete metric space whenever (X, d_X) is compact and (Y, d_Y) is complete).

(b) Let $C(\mathbb{R}, \mathbb{R})$ be the space of continuous functions from $\mathbb{R}$ to $\mathbb{R}$ and let

$$\boxed{C_0(\mathbb{R}, \mathbb{R}) := \left\{ f \in C(\mathbb{R}, \mathbb{R}) \colon \quad \lim_{|x|\to\infty} f(x) = 0 \right\}}\,.$$

(i) Show that $C_0(\mathbb{R}, \mathbb{R}) \subset \mathcal{B}(\mathbb{R}, \mathbb{R})$.

(ii) Show that $(C_0(\mathbb{R}, \mathbb{R}), \|\cdot\|_\infty)$ is a Banach space. (Hint: you can show that $C_0(\mathbb{R}, \mathbb{R})$ is a closed subspace of $\mathcal{B}(\mathbb{R}, \mathbb{R})$ and use (a).)

We will prove in the next lecture that a normed space $(V, \|\cdot\|)$ is a Banach space if and only if every absolutely converging series in $(V, \|\cdot\|)$ does converge.